Dr.-Ing. Danish **IQBAL**

Computational Engineer | Simulation Software Engineer

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10/03/1991

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PROFILE

Ph.D. in Computational Mechanics with seven years of research in numerical methods, multi-physics simulation, and fracture mechanics. Four peer-reviewed journal publications, international research collaboration, and multiple conference presentations. Complemented by hands-on industry experience in software development, data pipelines, API integration, and workflow automation. Focused on developing simulation software that integrates data management and AI/ML capabilities to solve complex multi-physics engineering problems.

PROFESSIONAL EXPERIENCE

07/2025 - present

Software Engineer

CleverBuy GmbH, Krefeld, Germany

- Developed automation systems replacing manual workflows with real-time, automated processes across multiple connected business systems.
- Built large-scale data pipelines integrating third-party platforms via APIs for continuous, automated data exchange and processing.
- Designed and implemented an asynchronous job engine using Celery for continuous rule-based evaluation over large volumes of operational data.
- Scoping an AI integration layer to connect a language model to structured business data for natural language querying and insights.

08/2024 - 06/2025

Professional Development

- Pursued self-directed learning in full-stack development, data science, and machine learning as part of a deliberate transition from academic research into industry engineering roles.

08/2017 - 03/2024

Research Assistant

University of Duisburg-Essen, Essen, Germany

- Developed and implemented the Scaled Boundary Finite Element Method (SBFEM) in MATLAB to simulate thermoelastic fracture in specialized engineering materials.
- Utilized ANSYS, Abaqus, Gmsh, and ParaView for advanced simulations, including scripting for automation and customization.
- Published four peer-reviewed papers in prestigious journals with over 138 cumulative citations.
- Conducted research in collaboration with international institutions and presented research at multiple international conferences.
- Taught graduate courses in Structural Dynamics and programming with ANSYS, Fortran, C++, and Python.
- Supervised three master's and one bachelor's thesis in computational mechanics.
- Managed an e-learning project to digitize course content and develop tools including virtual labs and inverted classroom formats.

PROFESSIONAL EXPERIENCE

01/2020 - 03/2020

Visiting Researcher

Federation University, Ballarat, Australia

- Collaborated with the University of Duisburg-Essen and Federation University under a prestigious Australia-Germany research grant, advancing SBFEM for modeling thermoelastic fracture in specialized engineering materials.

01/2017 - 03/2017

Student Assistant

University of Duisburg-Essen, Essen, Germany

- Implemented a polygon-based SBFEM in FORTRAN and integrated it into the open-source software FEAPpv.

06/2015 - 10/2015

Student Worker

Mercedes-Benz Group AG, Düsseldorf, Germany

- Executed spot welding with jigs and performed automotive assembly line tasks.

10/2014 - 12/2014

Student Worker

Mercedes-Benz Group AG, Rastatt, Germany

- Carried out production tasks on automotive assembly line.

12/2013 - 12/2013

Logistics Assistant

Amazon Deutschland Services GmbH, Rheinberg, Germany

- Scanned items and prepared them for packaging.

06/2011 - 07/2011

Internship

Karachi Tools, Dies & Moulds Center, Karachi, Pakistan

- Practical training in CAD/CAM and nitriding heat treatment.

CORE COMPETENCIES

- Multi-Physics Simulation
- Numerical Methods
- Scientific Computing
- Software Development
- Workflow Automation
- Project Management

SKILLS

Numerical Methods:	• SBFEM	• FEM	• XFEM	• CFD
Simulation Tools:	• ANSYS	• Abaqus	• Gmsh	• ParaView
Programming:	• MATLAB	• Python	• Fortran	• C++
Dev. Stack:	• Django	• PostgreSQL	• REST APIs	• Git

EDUCATION

08/2017 - 07/2024

Ph.D. Computational Mechanics

University of Duisburg-Essen, Germany

- Thesis: Development of scaled boundary polygon elements for coupled thermoelastic fracture modeling.
- Grade: Magna cum laude (1.0)

EDUCATION

10/2013 - 07/2017

M.Sc. Computational Mechanics

University of Duisburg-Essen, Germany

- Thesis: Development of polygon elements based on the scaled boundary finite element method.
- Grade: Good (2.0)

01/2009 - 12/2012

B.E. Automotive Engineering

NED University of Engineering & Technology, Pakistan

- Thesis: Design and analysis of motorcycle oil pump.
- Project: Design, development, and manufacturing of Formula Student car.
- Grade: First Division (2.5)

PUBLICATIONS

Peer-Reviewed Journal Papers

M. D. Iqbal, et al. "Transient thermoelastic fracture analysis of functionally graded materials using the scaled boundary finite element method." Theoretical and Applied Fracture Mechanics, vol. 127, Oct. 2023, p. 104056.

M. D. Iqbal, et al. "Thermoelastic fracture analysis of functionally graded materials using the scaled boundary finite element method." Engineering Fracture Mechanics, vol. 264, Apr. 2022, p. 108305.

M. D. Iqbal, et al. "Development of the scaled boundary finite element method for crack propagation modeling of elastic solids subjected to coupled thermo-mechanical loads." Computer Methods in Applied Mechanics and Engineering, vol. 387, Dec. 2021, p. 114106.

E.T. Ooi, *M. D. Iqbal*, et al. "A polygon scaled boundary finite element formulation for transient coupled thermoelastic fracture problems." Engineering Fracture Mechanics, vol. 240, Dec. 2020, p. 107300.

Conference Papers

C. Birk, R. Assaf, *M. D. Iqbal*, H. Gravenkamp, "Numerische strukturmodellierung mit polygon-elementen." Baustatik-Baupraxis 15 (2024).

M. D. Iqbal, et al., "Scaled boundary finite element modelling of thermally induced crack propagation." Proceedings in Applied Mathematics and Mechanics 19 (2019).

M. D. Iqbal, C. Birk, "Numerical modelling of thermally induced crack propagation in functionally graded materials using the scaled boundary finite element method." Forschungskolloquium 2018 Grasellenbach (2018).

Theses

M. D. Iqbal, "Development of scaled boundary polygon elements for coupled thermoelastic fracture modeling." PhD Thesis, University of Duisburg-Essen (2024).

M. D. Iqbal, "Development of polygon elements based on the scaled boundary finite element method." Master Thesis, University of Duisburg-Essen (2017).

LANGUAGES

Urdu (native) | English (business fluent) | German (fluent)

Essen, 9/4/2026